DHSNet: Dual Classification Head Self-Training Network for Cross-Scene Hyperspectral Image Classification

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***Abstract*—Due to the difficulty of obtaining labeled data for hyperspectral images (HSIs), cross-scene classification has emerged as a widely adopted approach in the remote sensing community. It involves training a model using labeled data from a source domain (SD) and unlabeled data from a target domain (TD), followed by inference on the TD. However, variations in the reflectance spectrum of the same object between the SD and the TD, as well as differences in the feature distribution of the same land cover class, pose significant challenges to the performance of cross-scene classification. To address this issue, we propose a dual classification head self-training network (DHSNet). This method aligns class-wise features across domains, ensuring that the trained classifier can accurately classify TD data of different classes. We introduce a dual classification head self-training strategy for the first time in the cross-scene HSI classification field and design a self-training loss based on the prediction of the two classification heads. The proposed approach mitigates the domain gap while preventing the accumulation of incorrect pseudo-labels in the model. In addition, we incorporate a novel central feature attention mechanism to enhance the model's capacity to learn scene-invariant features across domains. DHSNet significantly outperforms state-of-the-art methods on three cross-scene HSI datasets, achieving 80.23% ± 1.92% overall accuracy (OA) on Houston dataset. The code for DHSNet will be available at https://github.com/liurongwhm**

***Index Terms*—Central attention, cross-scene classification, domain adaptation (DA), hyperspectral image (HSI) , self-training.**

I. Introduction

With the advancements in sensor technology, aerospace engineering, and computer science, hyperspectral image (HSI) has become an important research area in remote sensing [1], [2], [3]. It is widely applied in various fields, such as agricultural monitoring [4], pollution assessment [5], environmental protection [6], and smart city development [7]. Owing to its high spectral resolution, HSI provides strong technical support for precise land cover classification. The goal of HSI classification is to assign each pixel to a predefined category [8], [9].

HSI classification faces significant challenges due to limited labels and high dimensionality [10]. Over the past few decades, HSI classification methods for the same scene have been extensively studied [11], [12], [13], [14], [15], [16], generally based on the assumption that the training and test data are independent and identically distributed. While this assumption roughly holds within the same scene, enabling these methods to achieve remarkable success, it breaks down in cross-scene scenarios [17]. Due to the labeling burden and high time cost [18], the HSI used for model training (source domain, SD) and the HSI to be classified (target domain, TD) are often from different scenes. As a result, applying models trained in one region to another region becomes crucial for practical applications, making cross-domain classification a key focus when classifying HSIs. Due to the variations in sensor conditions, geographical locations, and environmental factors, SD and TD exhibit significant differences in spectral reflectance and feature distribution, even for the same land cover class [19]. The above phenomenon results in a domain shift, where the distribution mismatch between the two domains leads to a substantial drop in classification performance.

To address the challenge of model transferability across different scenes, domain adaptation (DA) methods from transfer learning have been extensively developed. DA focuses on adapting models trained on the SD to perform well on a specific TD by leveraging available TD data, and in some cases, partial TD labels, with the ultimate goal of overcoming domain shift and transferring domain-invariant knowledge from the SD to the TD. Owing to their practicality and efficiency, DA-based approaches have gained increasing attention, especially in cross-scene HSI classification tasks.

Traditional DA methods can be categorized into two main types [20]: instance-based methods [21], [22], [23] and feature-based methods [24], [25], [26]. Instance-based methods focus on minimizing the domain discrepancy by reweighting the SD samples and training models on these

Received 4 April 2025; revised 19 August 2025 and 30 October 2025; accepted 15 November 2025. Date of publication 24 November 2025; date of current version 4 December 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62201622. *(Corresponding authors: Jiaqi Yang; Meiqi Hu.)*

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Digital Object Identifier 10.1109/TGRS.2025.3636101

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adjusted datasets. Feature-based methods, on the other hand, aim to construct a shared feature space where the distributions of SD and TD can be aligned. However, traditional DA methods face difficulties in feature construction and often fail to learn transferable features.

The ability of deep learning to extract high-level and abstract features has led to the flourishing of DA, which can be broadly categorized into discrepancy-based methods and adversarial-based methods [27]. Discrepancy-based methods explicitly quantify feature distribution differences using statistical metrics. A widely adopted metric is maximum mean discrepancy (MMD), which projects cross-domain data into a reproducing kernel Hilbert space (RKHS) and computes their distribution distance [28]. Variants include multikernel MMD [29] and joint MMD across layers [30]. Other discrepancy metrics include CORAL, which aligns domain-specific features by minimizing the covariance distance between the SD and TD [31], [32], and optimal transport (OT) theory, which geometrically compares distributions through mass transportation costs [33]. OT-based extensions further enhance alignment, such as DeepJDOT for joint distribution adaptation [34] and clustering-based OT (COT) integrating clustering operations [35]. In contrast, adversarial-based methods implicitly align features through domain-confusion objectives [36]. Inspired by GANs [37], domain-adversarial neural networks (DANNs) [38] employ a domain discriminator trained adversarially against feature extractors to erase domain-specific patterns. Beyond the use of a single domain discriminator, some studies explore alternative approaches. Methods such as [39], [40], and [41] employ two classifiers as discriminators to maximize the discrepancy between their predictions, while the feature extractor is simultaneously trained to minimize this discrepancy. This adversarial process helps bridge the gap between the SD and TD by making the learned features transferable across domains, effectively reducing domain shift.

Recently, deep DA methods have shown effectiveness in cross-scene HSI classification [42]. Liu et al. [43] proposed a method that aligns class-conditional distributions by combining class-wise DANN and MMD. Zhang et al. [44] proposed a topological structure and semantic information transfer network (TSTnet), which aligns both the statistical and geometric distributions between SD and TD. Zhao et al. [45] presented a framework toward multilevel features and decision boundaries (ToMF-B), which simultaneously aligns task-related features and learns task-specific decision boundaries. Li et al. [46] proposed a dual-channel residual network with a dynamic distribution adaptation strategy (DDAN-JCA). Feng et al. [47] proposed HyperEDL, a spectral–spatial evidence learning framework, to mitigate epistemic uncertainty in cross-scene HSI classification by adaptively aggregating multiorder contextual evidence. These methods incorporate the characteristics of hyperspectral imagery into DA model design, achieving notable results. Beyond task-specific adaptation networks, several recent works have developed general foundation models for HSI [48], [49], [50], which lay the groundwork for transferable and scalable feature representations in cross-scene applications.

Some methods focus on self-training, a semi-supervised technique that mitigates domain shift by iteratively assigning pseudo-labels to TD samples and retraining models with an expanded labeled dataset [51], [52]. For example, Fang et al. [53] developed a confident learning-based biclassifier adversarial neural network (CLDA) with a dynamic distribution adaptation strategy, dynamically adjusting the importance of marginal and conditional distributions. Feng et al. [54] proposed the pseudo-label-assisted subdomain adaptation (PASDA) method, which integrates subdomain alignment with a reweighted pseudo-label generation strategy to enhance domain-invariant feature learning and sample selection. However, pseudo-supervisory information often directly influences the optimization of overall model parameters, leading to incorrect judgments that degrade subsequent training and cause unpredictable performance drops. Some works [55], [56], [57] that introduce architectural innovations have demonstrated potential in mitigating these challenges, yet their applicability and efficacy in cross-scene HSI classification remain underexplored. Moreover, most existing methods for cross-scene HSI classification primarily focus on designing overall frameworks or DA strategies, often overlooking the critical role of feature extractors in effectively capturing domain-invariant representations. As a result, they may struggle to generalize across varying spatial scenes.

To overcome these limitations, we propose a dual classification head self-training network (DHSNet). This method introduces a novel central feature attention mechanism during the feature extraction stage and adopts class-wise feature alignment between the SD and TD. Furthermore, it uses dual classification heads for self-training, reducing model bias toward the SD while preventing the accumulation of incorrect pseudo-labels, thereby enhancing model transferability and cross-scene classification performance.

The main contributions of our proposed method are summarized as follows.

1) A unified cross-scene HSI classification framework is proposed. This framework consists of an effective feature extractor, class-wise feature alignment, and two classification heads. This complete pipeline is designed to jointly address domain shift and pseudo-label noise, thereby improving the generalization ability of the model across different scenes.
2) A central feature attention-aware convolution (CFAAC) block is proposed. It directly enhances attention to central features, which are critical for capturing cross-scene-invariant information. By focusing on these key features, the model's capacity to represent generalizable features is improved, making it well suited for cross-scene classification tasks.
3) To the best of our knowledge, this is the first work to employ a dual classification head self-training strategy in HSI cross-scene classification. This strategy mitigates the impact of incorrect pseudo-supervisory information from the TD, ensuring the effectiveness of self-training and the robustness of the overall model.

The remainder of this article is organized as follows. Section II reviews related works on DA and self-training

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techniques. Section III elucidates the overall framework and detailed implementation of the proposed method. Extensive experiments and their analyses are presented in Section IV. Finally, conclusions are drawn in Section V.

II. Related Works

Measuring the distance between two distributions is essential for aligning the SD and TD distributions. Gretton et al. [28] proposed a statistical method, MMD, to test whether two distributions, p and q, are different based on samples drawn from each. MMD quickly became one of the most widely used techniques for quantifying distribution differences between domains. The MMD distance is defined as

$$\mathrm{MMD}\left(\mathcal{F}, p, q\right) := \sup_{f\in\mathcal{F}} \left(\boldsymbol{E}_{\boldsymbol{x}^s\sim p}\left[f\left(\boldsymbol{x}^s\right)\right] - \boldsymbol{E}_{\boldsymbol{x}^t\sim q}\left[f\left(\boldsymbol{x}^t\right)\right]\right) \quad (1)$$

where $\mathcal{F}$ represents a class of functions $f : \mathbb{R}^d \to \mathbb{R}$, with d being the data dimension; and $\boldsymbol{x}^s$ and $\boldsymbol{x}^t$ denote data from the SD and TD, respectively. The operator $\sup(\cdot)$ denotes the supremum. Specifically, MMD is defined as the supremum of the difference between the expected values of the two distributions after mapping through functions in $\mathcal{F}$.

Given that only a limited number of samples are observed from the distributions, a biased empirical estimate of MMD is often more practical, which can be formulated as

$$\mathrm{MMD}_b\left(\mathcal{F}, \mathcal{D}_s, \mathcal{D}_t\right) := \sup_{f\in\mathcal{F}} \left(\frac{1}{n_s}\sum_{i=1}^{n_s} f\left(\boldsymbol{x}_i^s\right) - \frac{1}{n_t}\sum_{j=1}^{n_t} f\left(\boldsymbol{x}_j^t\right)\right) \quad (2)$$

where $\mathcal{D}_s$ and $\mathcal{D}_t$ represent the SD and TD, respectively. $\{\boldsymbol{x}_i^s\}_{i=1}^{n_s}$ and $\{\boldsymbol{x}_j^t\}_{j=1}^{n_t}$ are samples from $\mathcal{D}_s$ and $\mathcal{D}_t$, respectively.

If $\mathcal{F}$ is the unit ball in an RKHS, the empirical MMD can be efficiently computed as

$$\mathrm{MMD}\left(\mathcal{D}_s, \mathcal{D}_t\right) = \left\| \frac{1}{n_s}\sum_{i=1}^{n_s} \phi\left(\mathbf{x}_i^s\right) - \frac{1}{n_t}\sum_{j=1}^{n_t} \phi\left(\mathbf{x}_j^t\right) \right\|_{\mathcal{H}}^2 \quad (3)$$

where $\mathcal{H}$ is the RKHS endowed with a characteristic kernel and $\phi(\cdot)$ denotes the feature mapping that transforms the original samples into the RKHS. By square expansion of the formula, the inner product in the RKHS space can be converted into a kernel function, allowing the MMD to be calculated directly through the kernel function.

Building on the MMD metric, Ghifary et al. [58] introduced it into feedforward neural networks, pioneering the application of MMD in deep transfer learning. Then, this approach has been widely adopted. Zhu et al. [59] proposed the deep subdomain adaptation network (DSAN), which introduces the local MMD (LMMD) to better align feature distributions between the SD and TD by considering subdomain structures. LMMD can be formulated as

$$\mathrm{LMMD}\left(\mathcal{D}_s, \mathcal{D}_t\right) = \frac{1}{C}\sum_{c=1}^{C} \left\| \sum_{\mathbf{x}_i^s\in\mathcal{D}_s} w_i^{sc}\phi\left(\mathbf{x}_i^s\right) - \sum_{\mathbf{x}_j^t\in\mathcal{D}_t} w_j^{tc}\phi\left(\mathbf{x}_j^t\right) \right\|_{\mathcal{H}}^2 \quad (4)$$

where w_i^{sc} and w_j^{tc} denote the weights of the samples $\mathbf{x}_i^s$ and $\mathbf{x}_j^t$ belonging to class c, respectively; and C is the total number of categories, with the constraint that $\sum_{i=1}^{n_s} w_i^{sc} = 1$ and $\sum_{j=1}^{n_t} w_j^{tc} = 1$. The weights w_i^c for samples $\mathbf{x}_i$ from the SD and TD are computed as

$$w_i^{sc} = \frac{\boldsymbol{y}_{ic}^s}{\sum_{j=1}^{n_s} \boldsymbol{y}_{jc}^s} \quad (5)$$

$$w_i^{tc} = \frac{\boldsymbol{p}_{ic}^t}{\sum_{j=1}^{n_t} \boldsymbol{p}_{jc}^t} \quad (6)$$

where $\boldsymbol{y}^s$ is the one-hot vectors for samples from the SD and $\boldsymbol{p}^t$ is the probability vectors predicted by the model for samples from the TD.

III. Methodology

The flowchart of the proposed DHSNet is presented in Fig. 1, which consists of three main parts: feature extractor, DA, and self-training. The feature extractor processes samples from the SD and TD to extract transferable features. DA aligns these features at the class level across domains. The self-training part utilizes predictions for TD samples to further optimize the model. Through the collaboration of these three parts, DHSNet can efficiently learn cross-scene-invariant features while reducing the impact of erroneous pseudo-labels.

As shown in Fig. 1, patches from labeled SD and unlabeled TD images, denoted as $I \in \mathbb{R}^{\mathrm{ps}\times\mathrm{ps}\times c}$, where ps is the patch size and c is the number of spectral bands, are randomly selected (shown by black lines) and fed into the feature extractor. In the flowchart, purple lines represent labeled data flow, gray lines represent unlabeled data flow, and blue lines indicate the use of probability vectors. The extracted features from SD and TD are denoted as $\boldsymbol{z}^s$ and $\boldsymbol{z}^t$, respectively. These features are input into the classification head h_{cls} to get probability vectors $\boldsymbol{p}^s$ and $\boldsymbol{p}^t$, while the pseudo-head h_{psd} obtains only $\boldsymbol{z}^t$ and outputs predictions $\boldsymbol{p}^{t\prime}$. The DA part applies the LMMD loss function for class-wise feature alignment between SD and TD. During self-training, pseudo-labels from TD samples, generated by h_{cls}, are used to supervise the predictions from h_{psd}. Then, the parameters of both the feature extractor and h_{psd} are optimized through gradient backpropagation. During testing, the well-trained feature extractor and classification head are used to classify TD samples by extracting domain-invariant features.

A. Central Feature Attention-Aware Convolution

To ensure our model accurately captures and represents hyperspectral cross-scene-invariant features, we introduce a novel CFAAC module in the feature extractor. This module is motivated by the need to address spatial background noise variability and learning bias in the SD. Traditional CNNs often fail to generalize across scenes because they treat spatial neighbors equally and are sensitive to domain-specific interference. In contrast, our design leverages the observation that the center pixel within an image patch typically preserves the most stable and class-representative spectral signature of the target object. By focusing attention on these central features, the module learns to emphasize invariant,

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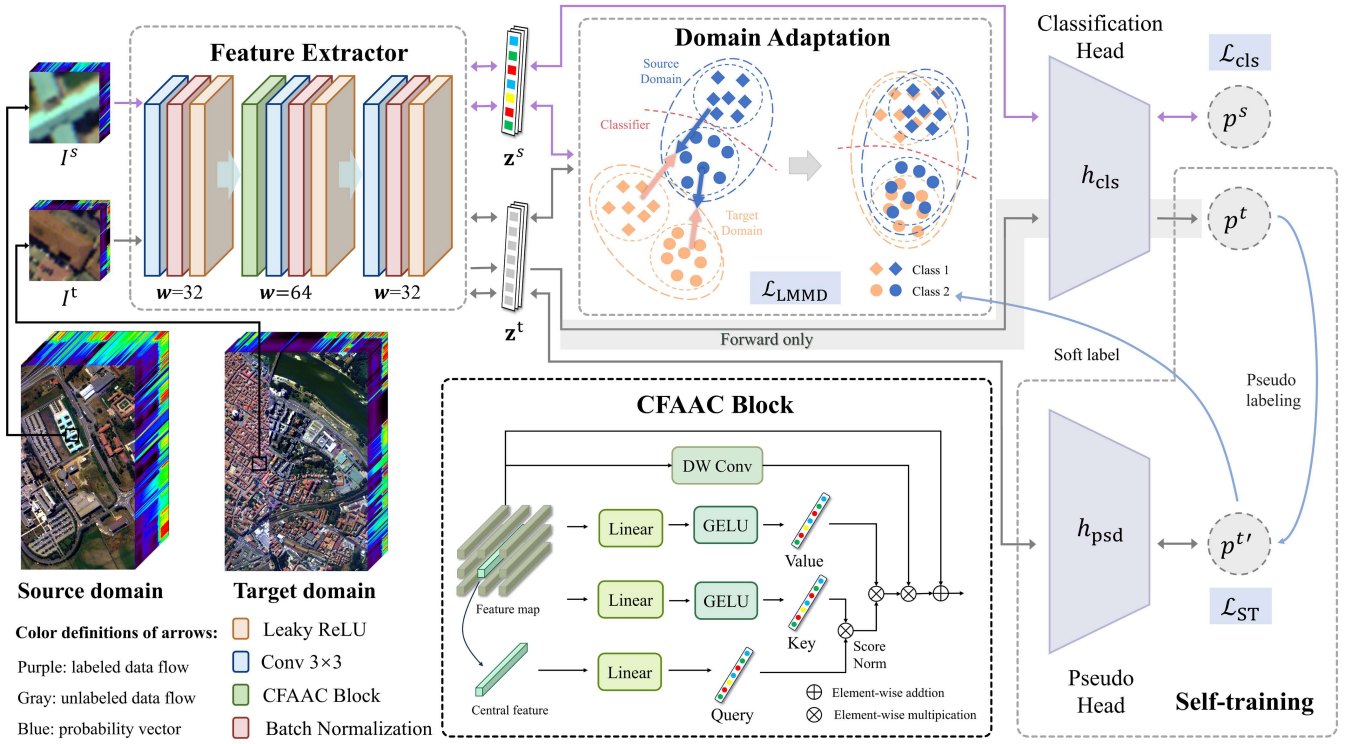


Fig. 1. Flowchart of the proposed DHSNet.

category-relevant information while suppressing scene-specific noise and peripheral distractions.

To implement this motivation, we design the CFAAC block to selectively enhance category-relevant spectral responses while reducing irrelevant spatial noise. Rather than statically emphasizing the center region, a gated mechanism is introduced to dynamically scale convolutional outputs at each spatial location based on their similarity to the central feature in latent space. This similarity-driven weighting enables the network to prioritize spatial regions whose spectral characteristics closely match the central pixel, thus suppressing distortions caused by SD spatial bias or scene-specific variation. Furthermore, the high-dimensional nature of latent features combined with nonlinear activation functions reinforces this adaptive filtering effect, enabling the model to better capture stable, transferable information for cross-scene classification.

The feature extractor consists of three units, with the channel dimensions for each unit represented by $\boldsymbol{w}$ in the flowchart (Fig. 1). The first and third units follow a sequential structure, comprising a 3×3 convolution layer, batch normalization, and the leaky rectified linear unit (ReLU) activation function [60]. The second unit integrates a CFAAC block before these layers. The channel dimension in the second unit is set to 64, which is twice that of the other units to ensure thorough feature extraction.

Details of the CFAAC block are also presented in Fig. 1. Inspired by the gated linear unit [61], the CFAAC block consists of two streams: central feature attention and convolution. For unit i, the input feature map $F_i \in \mathbb{R}^{\mathrm{ps}_i \times \mathrm{ps}_i \times w_i}$ is first fed into two separate linear layers, followed by the Gaussian error linear unit (GELU) activation function [62], to generate the key matrix $K_i \in \mathbb{R}^{(\mathrm{ps}_i \times \mathrm{ps}_i) \times w_i}$ and the value matrix $V_i \in \mathbb{R}^{(\mathrm{ps}_i \times \mathrm{ps}_i) \times w_i}$. Simultaneously, the central pixel of F_i is fed into another linear layer to generate a query vector $Q_i \in \mathbb{R}^{1 \times w_i}$. Each vector in K_i computes its similarity with Q_i via an inner product operation, determining the feature similarity in the latent space. These similarities are then normalized and applied as attention weights to each position. The attention matrix is multiplied by V_i to obtain the result of the central feature attention stream.

In the convolution stream, spatial information already utilized in the central feature attention is processed using a depth-wise convolution, grouped by the channel dimension w_i to enhance computational efficiency. Finally, the outputs from the two streams are combined by element-wise multiplication and added to the original feature map F_i using a residual design [63], producing the output of the CFAAC block. The specific formulations are as follows:

$$K_i = \mathrm{GELU}\left(L_{\mathrm{key}}\left(F_i\right)\right) \tag{7}$$

$$V_i = \mathrm{GELU}\left(L_{\mathrm{value}}\left(F_i\right)\right) \tag{8}$$

$$Q_i = L_{\mathrm{query}}\left(\mathrm{Center}\left(F_i\right)\right) \tag{9}$$

$$\mathrm{CFAAC}\left(F_i\right) = \frac{Q_i^T K_i}{\sqrt{\mathrm{ps}_i}} V_i \otimes \mathrm{DWConv}\left(F_i\right) + F_i \tag{10}$$

where $L(\cdot)$ represents a linear layer and Center$(\cdot)$ refers to the extraction of the central pixel. The operator $\otimes$ denotes the element-wise multiplication, and ps_i is the patch size of F_i. DWConv$(\cdot)$ represents the depth-wise convolution, a

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convolution operation where each input channel is convolved with a separate kernel.

The central feature attention mechanism differs from the scaled dot-product attention proposed by Vaswani et al. [64] by altering the many-to-many relationship between key and query vectors. While both works measure pixel similarity, in our design, each pixel queries only the central one. This dynamic approach modulates the convolution process by giving more attention to features closer to the center. Another key distinction is the use of activation functions in our approach, which enhances the model's nonlinear representational capacity and further differentiates our central feature attention from the scaled dot-product attention.

In summary, the core design purpose of the CFAAC module is to extract robust, discriminative, and domain-invariant features for cross-scene HSI classification. Specifically, by generating the query vector solely from the central pixel, a strong inductive bias is introduced that steers the attention mechanism toward semantically representative regions while suppressing background noise and domain-specific variations. Extended experiments demonstrate that applying a nonlinear GELU activation after the key and value projections significantly enhances the representational capacity of the features. The final output of the CFAAC block integrates the central feature attention mechanism, local spatial modeling via DWConv, and the stability benefits of residual connections, thereby effectively improving the model's adaptability to cross-scene variations and classification accuracy. In the overall framework, CFAAC provides more stable and domain-invariant feature representations, which make the feature distributions of SD and TD more consistent and thus facilitate subsequent domain alignment and self-training processes.

B. Domain Adaptation

LMMD distance is applied in the DA part of DHSNet to align the feature distributions from the SD and TD corresponding to the same land cover class. In practice, the Gaussian kernel, a type of radial basis function, is utilized as the kernel function in LMMD. Since the mapping function $\phi(\cdot)$ in (4) cannot be computed directly, we expand the squared term in the formula and exploit the properties of the kernel function in RKHS to derive the following expression:

$$\begin{aligned}\mathcal{L}_{\mathrm{LMMD}} = \frac{1}{C}\sum_{c=1}^{C}\Bigg[&\sum_{i=1}^{n_s}\sum_{j=1}^{n_s} w_i^{sc} w_j^{sc} r\left(\boldsymbol{z}_i^s, \boldsymbol{z}_j^s\right) \\ &+ \sum_{i=1}^{n_t}\sum_{j=1}^{n_t} w_i^{tc} w_j^{tc} r\left(\boldsymbol{z}_i^t, \boldsymbol{z}_j^t\right) \\ &-2\sum_{i=1}^{n_s}\sum_{j=1}^{n_t} w_i^{sc} w_j^{tc} r\left(\boldsymbol{z}_i^s, \boldsymbol{z}_j^t\right)\Bigg] \end{aligned} \tag{11}$$

where the kernel $r(\cdot,\cdot)$ represents the inner product of the features in RKHS. The computations of w_i^{sc} and w_i^{tc} follow the same procedures as those described in (5) and (6), respectively. Equation (11) is directly applied as the DA loss in DHSNet.

C. Self-Training

A key challenge in traditional self-training is error accumulation, where incorrect pseudo-labels generated by the model are used for retraining, leading to amplified biases. To address this, DHSNet introduces a dual head self-training strategy. The core philosophy of it is to fundamentally decouple the pseudo-label generation process from the utilization process. Specifically, two classification heads with identical architectures and different parameters are used independently: the classification head h_{cls}, which generates pseudo-labels, and the pseudo-classification head h_{psd}, which utilizes these labels.

As presented in the flowchart, h_{cls} takes inputs $\boldsymbol{z}^s$ and $\boldsymbol{z}^t$, representing features from SD and TD, respectively, and outputs the land cover class probability vectors $\boldsymbol{p}^s$ and $\boldsymbol{p}^t$. Here, $\boldsymbol{p}^s$, the classification result of the SD samples, is compared with the true labels, and the cross-entropy loss function is employed to optimize the parameters of both the feature extractor and the classification head h_{cls}.

The cross-entropy loss of a single sample is defined as

$$\mathcal{L}_{\mathrm{CE}}\left(\boldsymbol{p}, \boldsymbol{y}\right) = -\sum_{c=1}^{C} \boldsymbol{y}_c \log \boldsymbol{p}_c \tag{12}$$

where $\boldsymbol{y}$ is the one-hot encoded label vector, $\boldsymbol{p}$ denotes the probabilistic prediction vector, and C is the number of classes. The classification loss for the SD is then defined as

$$\mathcal{L}_{\mathrm{cls}} = \frac{1}{n_s}\sum_{i=1}^{n_s} \mathcal{L}_{\mathrm{CE}}\left(\boldsymbol{p}_i^s, \boldsymbol{y}_i^s\right) \tag{13}$$

where n_s is the number of samples from the SD.

The probabilistic prediction $\boldsymbol{p}^t$ is converted into a one-hot pseudo-label by the argmax function and then assigned to TD samples whose maximum predicted probability exceeds a confidence threshold τ (empirically, set to 0.95 in this article). h_{psd} processes only $\boldsymbol{z}^t$, and outputs their probabilistic predictions $\boldsymbol{p}^{t\prime}$, which are used for self-training. The self-training loss for TD samples is formulated as

$$\mathcal{L}_{\mathrm{ST}} = \frac{1}{n_t}\sum_{i=1}^{n_t} \mathbb{I}\left(\max\left(\boldsymbol{p}_i^t\right) > \tau\right) \cdot \mathcal{L}_{\mathrm{CE}}\left(\boldsymbol{p}_i^{t\prime}, \mathrm{onehot}\left(\boldsymbol{p}_i^t\right)\right) \tag{14}$$

where n_t is the number of samples from the TD. $\mathrm{I}(\cdot)$ is the indicator function that outputs 1 if the condition is satisfied and 0 otherwise.

For h_{cls}, through its simultaneous supervision by labeled SD data and interaction with domain alignment, the pseudo-labels it generates inherently carry a perception of the domain shift. In this scenario, the role of h_{psd} transcends simple noise reduction in general semi-supervised learning; it becomes an independent learner specifically tasked with robustly learning more transferable feature representations from potentially compounded-biased pseudo-labels generated by h_{cls} on unlabeled data within the presence of domain shift. This independence ensures that pseudo-label biases stemming from the main classification head's SD influence do not directly and catastrophically contaminate the feature extractor.

In summary, the classification loss is used to optimize only the main classification head, while the self-training loss

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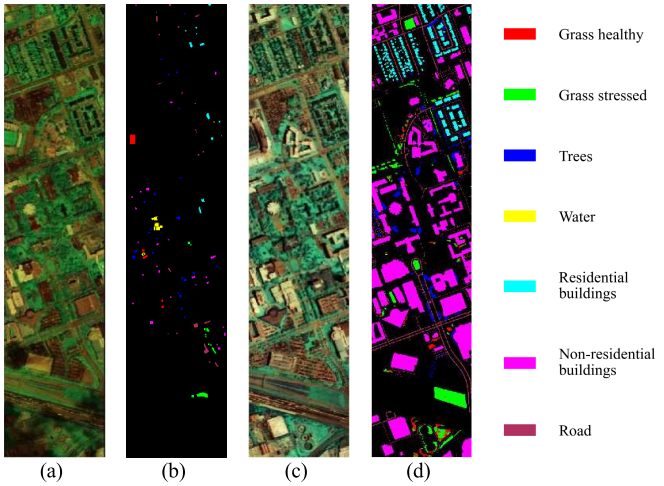


Fig. 2. Pseudo-color image and ground-truth map of Houston. (a) Pseudo-color image of Houston 2013. (b) Ground-truth map of Houston 2013. (c) Pseudo-color image of Houston 2018. (d) Ground-truth map of Houston 2018.

updates only the pseudo-label head. Although the pseudo-labels are generated based on the hard predictions from the main head, gradient backpropagation is explicitly truncated during self-training to prevent feedback from the pseudo-head from affecting the main classification head. During training, the main classification head maintains reliable discrimination of the SD, while the pseudo-classification head guides the shared features of TD samples to align with the source distribution through pseudo-label-driven optimization. This collaborative process effectively reduces the model's bias toward the SD and mitigates the accumulation of pseudo-label errors.

Integrating the above loss functions, the total training loss function of DHSNet is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \lambda_{\text{LMMD}}\mathcal{L}_{\text{LMMD}} + \lambda_{\text{ST}}\mathcal{L}_{\text{ST}} \tag{15}$$

where λ_{LMMD} and λ_{ST} are hyperparameters controlling the relative contributions of DA and self-training, respectively. Overall, the loss function of DHSNet consists of three parts. The classification loss for SD ensures that the model learns meaningful features for downstream tasks. The LMMD loss promotes class-wise feature alignment between SD and TD. The self-training loss extends the training data by incorporating pseudo-labeled TD samples, mitigating the model's bias toward SD.

IV. Experiment

A. Dataset

To evaluate the effectiveness of the proposed DHSNet, three publicly available cross-scene HSI datasets are used, including Houston, HyRANK, and Pavia.

1) Houston: This dataset was obtained from the University of Houston campus and the neighboring urban area. It consists of two scenes, Houston 2013 [65] (used as SD) and Houston 2018 [66] (used as TD), which were captured with different sensors and at different times in Houston, TX, USA. Houston 2013 includes 349 × 1905 pixels and 144 spectral bands, whereas Houston 2018 includes 209 × 955 pixels and 48 spectral bands. To align with Houston 2018 dataset, an overlapped region from Houston 2013 with 48 spectral bands and 209 × 955 pixels was selected. Both scenes share seven common land cover classes, with details shown in Table I and Fig. 2.

TABLE I
Land Cover Classes and the Numbers of Samples in Houston Dataset

No.	Class	Houston2013	Houston2018
C1	Grass healthy	345	1353
C2	Grass stressed	365	488
C3	Trees	365	2766
C4	Water	285	22
C5	Residential buildings	319	5347
C6	Non-residential buildings	408	32459
C7	Road	443	6365
	Total	2530	53200

TABLE II
Land Cover Classes and the Numbers of Samples in the HyRANK Dataset

No.	Class	Dioni	Loukia
C1	Dense Urban Fabric	1262	206
C2	Mineral Extraction Sites	204	54
C3	Non lrrigated Arable Land	614	426
C4	Fruit Trees	150	79
C5	Olive Groves	1768	1107
C6	Coniferous Forest	361	422
C7	Dense Sclerophyllous Vegetation	5035	2996
C8	Sparce Sclerophyllous Vegetation	6374	2361
C9	Sparsely Vegetated Areas	1754	399
C10	Rocks and Sand	492	453
C11	Water	1612	1393
C12	Coastal Water	398	421
	Total	20024	10317

2) HyRANK: This dataset is composed of two regions, Dioni (used as SD) and Loukia (used as TD). They are obtained from the Hyperion sensor (EO-1, USGS), featuring 176 spectral bands [67]. The sizes of Dioni and Loukia are 250 × 1376 and 249 × 945, respectively. There are 12 common classes between the two scenes, as detailed in Table II. The pseudo-color images and ground-truth maps for both scenes are shown in Fig. 3.

3) Pavia: This dataset includes Pavia University (PU, used as SD) [68] and Pavia Center (PC, used as TD) [69]. Both datasets were obtained using the Reflective Optics System Imaging Spectrometer (ROSIS) sensors in Pavia, Northern Italy. PU contains 610 × 610 pixels and 103 spectral bands, whereas PC consists of 1096 × 1096 pixels and 102 spectral bands. For experimental purposes, PU and PC were cropped to 610 × 315 pixels and 1096 × 715 pixels, respectively. To ensure consistency, the last spectral band of PU was removed to match the number of spectral bands with PC. Both datasets feature seven common land cover classes, with detailed sample

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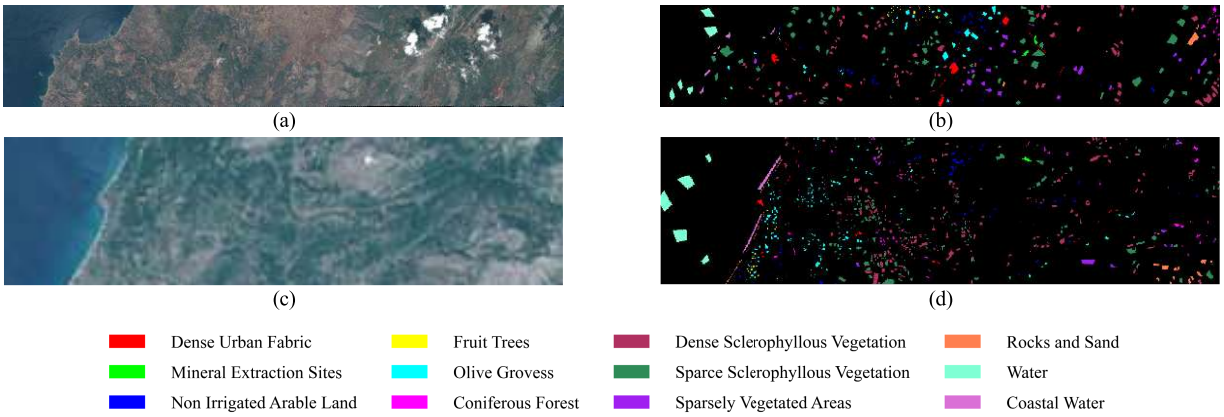


Fig. 3. Pseudo-color image and ground-truth map of HyRANK. (a) Pseudo-color image of Dioni. (b) Ground-truth map of Dioni. (c) Pseudo-color image of Loukia. (d) Ground-truth map of Loukia.

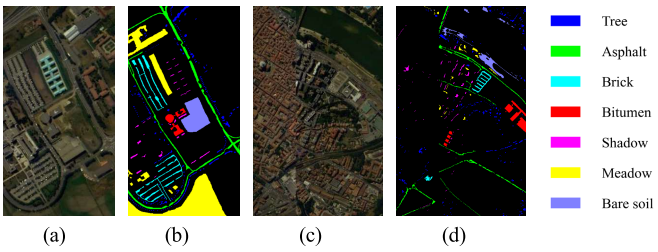


Fig. 4. Pseudo-color image and ground-truth map of Pavia. (a) Pseudo-color image of Pavia University. (b) Ground-truth map of Pavia University. (c) Pseudo-color image of Pavia Center. (d) Ground-truth map of Pavia Center.

TABLE III
LAND COVER CLASSES AND THE NUMBER OF SAMPLES IN PAVIA DATASET

No.	Class	Pavia University	Pavia Center
C1	Trees	3064	7598
C2	Asphalt	6631	9248
C3	Bricks	3682	2685
C4	Bitumen	1330	7287
C5	Shadow	947	2863
C6	Meadows	18649	3090
C7	Bare soil	5029	6584
Total		39332	39355

information in Table III. Fig. 4 displays their pseudo-color images and ground-truth maps.

B. Implementation Details

Several comparative methods are employed to illustrate the effectiveness of our experiment, including SpectralFormer [70] and state-of-the-art unsupervised DA algorithms: dynamic adversarial adaptation network (DAAN) [71], DSAN [59], TSTnet [44], CLDA [53], DDAN-JCA [46], and MSDA [72].

(1) For SpectralFormer, we adopt a patch-wise input strategy with a patch size of 7 × 7 and the number of neighboring bands set to 5 for all three datasets. The model is trained for 200 epochs with a learning rate of 0.005, optimized using the Adam optimizer. A StepLR scheduler is applied with a step size of one-tenth of the total epochs and a decay factor γ of 0.9.

(2) For DAAN, the training consists of 200 epochs with a batch size of 40. The learning rate is defined as $\mathrm{lr}_w = (\mathrm{lr}_0/(1+\alpha w)^\beta)$, where $\alpha = 10, \beta = 0.75$, and $\mathrm{lr}_0 = 0.01$. The optimizer is stochastic gradient descent (SGD) with a momentum of 0.9 and a weight decay of 0.0005. For all datasets, the input patch size is 5 × 5.

(3) For DSAN, the training consists of 300 epochs with a batch size of 100, and the learning rate follows the same decay strategy as DAAN. The optimizer is SGD with a momentum of 0.9 and a weight decay of 0.0001. For all datasets, the input patch size is 5 × 5, and the parameter λ is set to 0.5.

(4) For TSTnet, the training consists of 500 epochs with a batch size of 100. The learning rate decay strategy and optimizer settings are the same as those of DAAN. For all datasets, the input patch size is 13 × 13, with λ_1 and λ_2 set to 1 and 10 for Houston, respectively, and 1 and 0.1 for HyRANK and Pavia, respectively.

(5) For CLDA, the training consists of 100 epochs with a batch size of 36. lr_0 is 0.0001, with $\alpha = 0.1, \beta = 0.01$, and $\gamma = 0.01$. All other settings are consistent with the method's open-source code. The input patch size is 5 × 5 for all datasets.

(6) For DDAN-JCA, the training consists of 100 epochs with a batch size of 64. The parameter α is set to 0.5 for Pavia and 1 for Houston and HyRANK. The input patch size is 9 × 9 for Pavia and Houston, and 5 × 5 for HyRANK.

(7) For MSDA, the training lasts for 50 epochs with a batch size of 32 and a learning rate of 0.1. The temperature parameter T is set to 9 for Houston and Pavia, and 7 for HyRANK. The mask ratio r is 0.1 for Houston, and 0.7 for both Pavia and HyRANK. The input patch sizes are 7 × 7 for Houston, 9 × 9 for Pavia, and 3 × 3 for HyRANK.

The experimental setup for DHSNet utilizes the PyTorch framework, employing minibatch SGD for optimization. The momentum is set to 0.9, and l_2-norm regularization is set

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to 0.0001. The learning rate and batch size for all three datasets are set to 0.01 and 100. For the feature extractor, the convolution layers and batch normalization layers use the Kaiming distribution and are initialized with a constant value of 1. The learning rate adjustment follows the formula: $\mathrm{lr}_w = (\mathrm{lr}_0/(1+\alpha w)^{\beta})$, where w indicates the training progress linearly ranging from 0 to 1, lr_0 is the initial learning rate, $\alpha = 10$, and $\beta = 0.75$. The number of training epochs is 200. Patch sizes for Houston, HyRANK, and Pavia are empirically set to 15×15, 7×7, and 9×9, respectively. For all subsequent experiments, both DHSNet and the comparison methods are trained using 200 labeled samples per class from the SD and 200 unlabeled samples from the TD. The classification performance is quantitatively evaluated using the overall accuracy (OA), the average accuracy (AA), and the Kappa coefficient [73]. Their mathematical formulations are given as follows:

$$\mathrm{OA} = \frac{\sum_{i=1}^{C} n_{ii}}{N} \tag{16}$$

$$\mathrm{AA} = \frac{1}{C}\sum_{i=1}^{C}\frac{n_{ii}}{n_{i\cdot}} \tag{17}$$

$$\mathrm{Kappa} = \frac{P_o - P_e}{1 - P_e},\ \text{where} P_o = \mathrm{OA}, P_e = \frac{\sum_{i=1}^{C} n_{i\cdot} n_{\cdot i}}{N^2} \tag{18}$$

where n_{ii} is the number of correctly predicted samples for class i, $n_{i\cdot}$ is the total number of ground-truth samples in class i, $n_{\cdot i}$ is the total number of samples predicted as class i, N is the total number of test samples, and C is the number of classes.

To ensure unbiased results, all experiments are repeated five times independently, with the mean values of the evaluation metrics reported for comprehensive analysis.

C. Experimental Results

Tables IV–VI display the class-specific classification accuracy, OA, AA, and Kappa coefficients for the aforementioned methods across three datasets, respectively. Bold indicates the highest accuracy, and underline indicates the second-best accuracy. Visual classification results are further shown in Figs. 5–7. In these maps, labeled pixels are displayed as ground-truth and unlabeled pixels as background. From the quantitative results, we can see that the following conditions hold.

1) *Outstanding Quantitative Performance:* Compared to other methods, the proposed DHSNet achieves the best performance in terms of class-specific accuracy, OA, AA, and Kappa coefficient. For Houston dataset, DHSNet's OA exceeds 80.00%, outperforming TSTnet, which has high performance on this dataset with 78.84%, and leading by a large margin over all other methods. For the HyRANK dataset, where vegetation classes are prone to misclassification and most methods have an OA just above 60.00%, DHSNet's OA surpasses 70%. For Pavia dataset, DHSNet slightly outperforms CLDA (89.95%) and MSDA (90.65%), both of which perform well on this dataset, and substantially outperforms the rest of the methods.
2) *Excellent Class Balance:* Through class-wise feature alignment, DHSNet can obtain superior and balanced class-specific accuracy for some difficult-to-distinguish categories. For instance, in Houston dataset, DHSNet achieves nearly equal classification accuracy (about 80%) for both the first class (grass healthy) and the second class (grass stressed), indicating minimal class imbalance. In the HyRANK dataset, DHSNet significantly outperforms other methods in classifying olive groves, a class in which other methods achieve less than 25% accuracy. DHSNet achieves an average accuracy of 47.90% for this class, though with some room for improvement in stability.
3) Among pseudo-label-based methods, our approach achieves notable performance improvements. As shown in Table IV for Houston dataset, the OA of DHSNet is approximately 15% higher than that of CLDA. One possible reason is that CLDA is more sensitive to the number of TD samples: its confident learning mechanism relies on sufficient target data to select reliable pseudo-labels, and these pseudo-labels are directly used to optimize both feature extractors, making the model more susceptible to accumulated noise. In contrast, DHSNet mitigates this issue by employing a dual classification head strategy.
4) Deep transfer learning baseline models like DAAN and DSAN, not specifically designed for HSI, perform reasonably well across the datasets, particularly in cases with many classes (such as the HyRANK dataset). However, these methods reach a performance ceiling compared to HSI-specific methods, which incorporate additional mechanisms such as pseudo-labels or topological information.

D. Ablation Study

The ablation study is designed to demonstrate the validity of the CFAAC block, DA, and dual classification head self-training, and to evaluate their contributions. We adopt seven methods across three datasets. To verify the effectiveness of the CFAAC block, a baseline model using the same feature extractor architecture as DHSNet but without the CFAAC block is established. In addition, only the feature extractor of DHSNet is retained for this comparison. Furthermore, we separately add the DA and self-training components to the feature extractor to validate their respective effectiveness. To demonstrate the validity of the pseudo-classification head, a series of experiments using self-training, both with and without the pseudo-classification head, was implemented under various experimental conditions. For models without a pseudo-head, the training process consists of two stages: 1) initial supervised training on the SD labeled data for 20 epochs and 2) self-training using TD samples with hard pseudo-labels generated by the model itself while continuing to train on the SD data. It is noteworthy that all these experiments use the same feature extractor as DHSNet and the same pseudo-label criterion threshold τ. Specifically, we set $\mathrm{lr}_0 = 0.01$, the sample number of SD and TD to 200, $\lambda_{\mathrm{LMMD}} = 1$, $\lambda_{\mathrm{ST}} = 1$, and patch size to 13 for Houston dataset. For HyRANK and Pavia, patch

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TABLE IV
QUANTITATIVE CLASSIFICATION RESULTS ON HOUSTON DATASET (TD: HOUSTON 2018)

Class	SpectralFormer	DAAN	DSAN	TSTnet	CLDA	DDAN-JCA	MSDA	DHSNet
Grass healthy	64.21±4.82	58.05±23.59	50.38±34.86	60.64±27.58	71.25±13.52	49.24±33.84	**86.24±21.44**	84.12±9.29
Grass stressed	**93.09±4.32**	81.84±1.89	73.04±16.88	69.50±5.70	86.26±4.76	15.07±19.00	55.12±14.10	79.23±2.93
Trees	61.11±4.81	64.39±4.97	63.22±12.16	65.63±3.16	67.52±4.79	56.96±15.55	65.62±3.28	**71.76±2.82**
Water	**100.00±0.00**	91.82±8.33	91.82±4.45	**100.00±0.00**	85.45±11.28	43.64±18.32	82.73±9.05	96.36±7.27
Residential buildings	62.42±8.25	77.26±9.58	63.76±12.69	87.33±3.37	93.16±4.17	76.45±18.58	**93.23±1.37**	85.32±5.65
Non-residential buildings	73.08±5.79	64.08±2.91	67.41±6.78	**83.84±3.86**	54.26± 7.02	73.61±10.25	71.29±2.97	83.80±4.32
Road	61.93±5.20	57.83±15.77	52.77±9.17	62.91±7.99	73.47±6.79	27.28±34.25	**78.11±3.55**	61.33±7.84
OA (%)	71.68±2.73	66.16±3.90	65.16±4.14	78.84±2.61	64.55±4.22	61.48±1.40	72.91±1.63	**80.23±1.92**
AA (%)	73.69±2.04	70.75±4.40	66.05±5.88	75.69±5.19	75.91± 2.54	48.89±11.41	76.05±1.98	**80.27±1.41**
Kappa × 100	56.21±2.94	50.29±5.38	47.47±4.49	65.87±3.81	51.65±4.27	35.88±9.21	59.98±1.96	**68.52±2.07**

TABLE V
QUANTITATIVE CLASSIFICATION RESULTS ON THE HYRANK DATASET (TD: LOUKIA)

Class	SpectralFormer	DAAN	DSAN	TSTnet	CLDA	DDAN-JCA	MSDA	DHSNet
Dense Urban Fabric	1.94±2.83	30.29±7.43	28.54±17.53	43.69±22.73	28.93±27.35	6.70±1.12	26.89±9.68	**64.37±3.08**
Mineral Extraction Sites	7.78±13.73	**87.04±9.59**	81.11±19.68	38.89±47.66	79.63±39.82	30.37±37.27	84.81±15.73	75.56±11.26
Non lrrigated Arable Land	44.27±16.32	37.32±13.09	65.12±22.93	47.93±16.98	82.35±6.34	56.95±15.74	**88.62±2.31**	75.49±9.47
Fruit Trees	6.33±12.66	73.16±27.56	60.51±28.52	75.70±12.05	**99.75±0.51**	67.85±12.92	59.62±16.67	81.01±4.67
Olive Groves	0.70±1.41	17.16±16.30	23.43±27.51	9.65±12.50	3.41±2.97	5.35±3.22	**75.42±5.05**	47.90±16.70
Coniferous Forest	**60.19±16.74**	51.71±17.34	36.73±27.87	25.92±13.08	53.79±5.91	47.87±20.35	1.73±0.61	54.27±9.06
Dense Sclerophyllous Vegetation	65.95±17.35	67.46±6.86	61.79±3.94	71.68±7.50	66.56±4.85	63.55±9.26	43.55±10.94	**75.55±5.42**
Sparce Sclerophyllous Vegetation	56.05±12.10	49.14±14.24	56.73±14.39	59.88±11.29	21.75±11.07	24.37±5.63	64.81±6.37	**67.62±6.90**
Sparsely Vegetated Areas	14.24±9.15	29.42±20.12	46.42±28.68	50.03±21.62	45.31±24.72	22.81±9.90	33.21±10.16	**77.99±6.14**
Rocks and Sand	0.26±0.22	24.15±21.43	33.16±11.65	23.58±24.70	53.95±27.16	50.29±23.75	9.14±22.22	**57.00±21.77**
Water	80.89±16.14	97.47±2.61	94.93±8.11	99.77±0.46	**100.00±0.00**	95.03±6.80	**100.00±0.00**	**100.00±0.00.**
Coastal Water	8.17±4.28	98.19±2.21	77.58±37.77	**100.00±0.00**	**100.00±0.00**	99.95±0.10	90.00±30.00	99.71±0.38
OA (%)	48.29±8.24	57.32±3.74	58.32±5.13	60.77±2.03	53.73±2.47	49.73±3.81	59.60±5.52	**73.28±2.28**
AA (%)	28.90±3.47	55.21±2.50	55.50±6.25	53.89±4.76	61.29±5.40	47.59±3.90	56.48±4.24	**73.04±1.14**
Kappa × 100	37.88±8.19	49.75±3.98	50.75±6.00	52.54±1.84	47.23±-2.70	41.86±3.56	51.94±6.37	**68.04±2.46**

sizes of 7 and 9 are used, while other hyperparameters remain unchanged.

As shown in Table VII, the FE with the CFAAC block achieves approximately a 0.70% improvement in OA across the three datasets compared to the baseline. This demonstrates the effectiveness of the CFAAC block in extracting and representing domain-invariant features for HSI. Except for Houston dataset, incorporating only the LMMD loss significantly enhances the OA, with the HyRANK dataset showing an improvement of approximately 6.00% and Pavia dataset showing about 10.00%. Applying the self-training strategy alone results in an OA improvement of around 1% for Houston and HyRANK datasets, while Pavia dataset sees an increase of over 8.00%. Furthermore, the combined application of LMMD and self-training achieves the best performance across all datasets, outperforming the individual application of each component. This indicates that the combination of LMMD and self-training is effective for cross-scene HSI classification.

To validate the effectiveness of the pseudo-classification head, we observe that across all datasets, the pseudo-classification head self-training strategy (dual head groups) consistently outperforms that without it (single head groups), regardless of whether LMMD is applied. This improvement is particularly pronounced for Pavia dataset, followed

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TABLE VI
QUANTITATIVE CLASSIFICATION RESULTS ON PAVIA DATASET (TD: PAVIA CENTER)

Class	SpectralFormer	DAAN	DSAN	TSTnet	CLDA	DDAN-JCA	MSDA	DHSNet
Trees	84.70±5.89	89.60±11.43	87.72±8.48	87.77±8.06	94.95±3.41	**97.66±1.48**	93.36±4.58	94.81±2.96
Asphalt	76.16±5.94	82.64±11.36	87.94±2.60	92.55±3.82	**99.86±0.10**	92.54±2.24	98.23± 1.23	97.13±0.82
Bricks	72.01±12.27	**87.99±7.29**	80.53±21.40	75.86±6.94	76.60±3.85	77.91±6.60	48.80±31.43	84.27±5.43
Bitumen	74.11±7.07	61.61±28.87	65.57±3.71	0.97±0.34	71.80±24.43	22.14±21.66	82.52±1.32	**86.41±1.83**
Shadow	80.96±4.06	80.27±6.29	89.05±6.12	97.23±1.37	99.99±0.01	99.66±0.53	99.24±1.01	**100.00±0.00**
Meadows	81.69±2.13	47.92±15.28	65.18±19.13	77.18±9.70	85.88±9.87	73.37±12.97	**95.82±3.43**	84.76±6.97
Bare soil	88.65±5.48	78.32±8.60	77.81±5.21	86.65±5.41	92.72±2.66	78.30±8.00	**96.80±3.23**	91.09±1.37
OA (%)	80.02±1.82	76.83±4.92	79.85±2.48	71.68±1.45	89.85±4.62	76.13±3.95	90.65±2.89	**92.05±0.31**
AA (%)	79.75±1.22	75.48±3.44	79.11±3.41	74.03±1.98	88.83±3.85	77.37±2.15	87.82±4.96	**91.21±0.97**
Kappa × 100	76.09±2.10	72.13±5.87	75.76±3.02	66.64±1.69	87.72±5.65	71.16±4.79	88.73±3.50	**90.42±0.38**

TABLE VII
COMPREHENSIVE ABLATION STUDY OF DHSNET MODULES, INCLUDING PSEUDO-CLASSIFICATION HEAD

Method		baseline	FE only	FE+LMMD	FE+ST (single head)	FE+ST (dual head)	FE+LMMD+ST (single head)	FE+LMMD+ST (dual head)
Houston	OA(%)	77.86±2.10	78.54±1.24	76.45±0.93	78.11±2.30	79.17±1.82	78.12±1.87	**79.25±2.07**
	AA(%)	74.87±1.39	76.92±2.25	73.76±2.79	75.13±5.18	73.86±1.20	**78.76±3.24**	78.14±4.12
	Kappa×100	62.85±3.45	65.35±1.42	61.89±2.21	63.72±4.77	64.64±2.31	65.20±3.77	**67.02±3.65**
HyRANK	OA(%)	63.52±2.17	64.13±4.06	70.56±1.85	64.77±1.99	65.30±2.02	71.22±2.00	**71.25±1.11**
	AA(%)	58.99±2.83	56.30±1.86	72.58±2.48	61.49±2.86	63.42±4.05	71.65±3.93	**72.90±1.91**
	Kappa×100	55.96±2.29	56.82±4.48	64.80±1.89	57.72±2.43	58.31±2.38	65.68±2.11	**65.74±1.31**
Pavia	OA(%)	81.02±2.47	81.72±3.33	91.34±0.54	87.04±1.10	89.84±1.67	91.29±0.96	**92.05±0.31**
	AA(%)	80.64±1.83	79.81±6.46	90.29±0.28	85.62±0.99	87.57±1.45	90.43±1.35	**91.21±0.97**
	Kappa×100	77.02±3.00	77.89±4.07	89.57±0.65	84.40±1.32	87.73±2.00	89.51±1.15	**90.42±0.38**

FE: Feature Extractor; ST: Self-training

by the HyRANK dataset. Therefore, the effectiveness of the pseudo-classification head in self-training is successfully demonstrated.

E. Parameter Sensitivity Analysis

There are three parameters in DHSNet, i.e., patch size, λ_{LMMD}, and λ_{ST}. Patch size controls the size of the input HSI data cube, while λ_{LMMD} and λ_{ST} determine the weight of LMMD and self-training in the total loss function, respectively. We initially empirically set both λ_{LMMD} and λ_{ST} to 1 and searched for the optimal patch size for each dataset. Patches are selected from {5 × 5, 7 × 7, 9 × 9, 11 × 11, 13 × 13, 15 × 15, 17 × 17} for verification.

As shown in Fig. 8, the best patch sizes of Houston, HyRANK, and Pavia are 15, 7, and 11, respectively. The HyRANK dataset is most sensitive to patch size, showing a significant increase in OA of over 30% from 5 × 5 to 7 × 7, followed by a gradual decline as the patch size further increases; Pavia dataset is the least sensitive to patch size, with the variation in OA not exceeding 2% across different patch sizes.

In the following discussion, the patch size is set to the optimal value for the corresponding dataset. The value ranges of λ_{LMMD} and λ_{ST} are both set to {0.2, 0.4, 0.6, 0.8, 1}. Fig. 9 presents the changing trend of OA in the three datasets with different λ_{LMMD} and λ_{ST} combinations. As shown in Fig. 9, DHSNet is more sensitive to λ_{LMMD} than λ_{ST}. This is consistent with the contributions to the model's performance revealed by the ablation study when considering each part separately. The OA of Houston dataset generally decreases with the increase of λ_{LMMD}, while that of Pavia shows the opposite trend. HyRANK exhibits an initially increasing, then decreasing trend for λ_{LMMD}, and it is more volatile for λ_{ST}. Overall, the sensitivity to these two parameters, from least to most, is Pavia, Houston, and HyRANK.

Based on our experiments, the optimal λ_{LMMD} and λ_{ST} combinations for Houston, HyRANK, and Pavia are {0.2, 0.2}, {0.6, 0.4}, and {1, 0.8}, respectively. Notably, the discrepancies in OA obtained from these diverse parameter configurations are predominantly constrained within a 2% interval, substantiating the stability and robustness of our proposed model.

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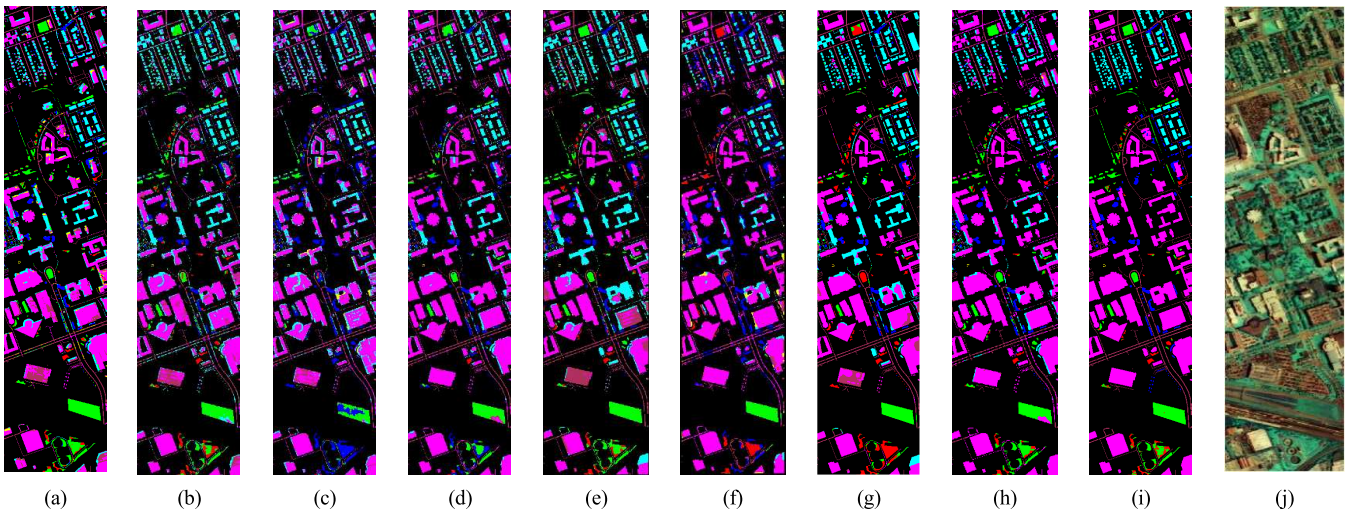


Fig. 5. Visualization of classification maps by different methods on Houston dataset, including: (a) SpectralFormer (70.87%), (b) DAAN (68.73%), (c) DSAN (64.11%), (d) TSTnet (79.64%), (e) CLDA (69.06%), (f) DDAN-JCA (67.02%), (g) MSDA (75.29%), (h) DHSNet (81.27%), (i) ground truth, and (j) pseudo-color image of Houston 2018.

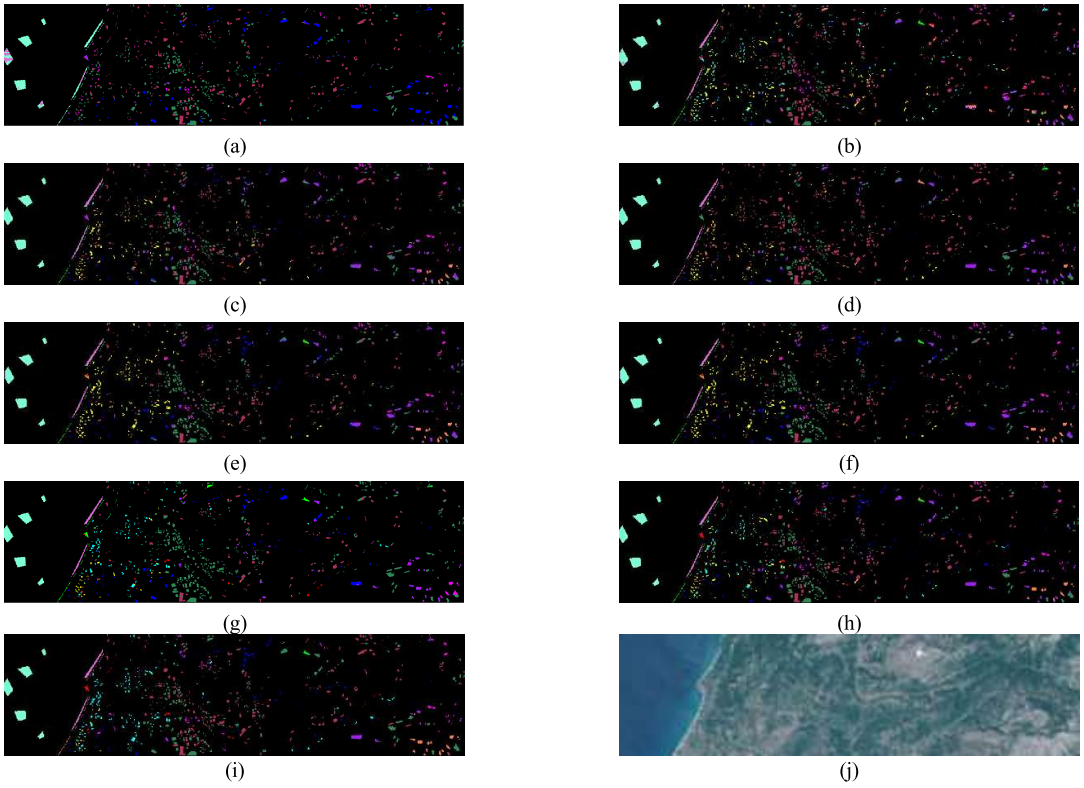


Fig. 6. Visualization of classification maps by different methods on the HyRANK dataset, including: (a) SpectralFormer (52.08%), (b) DAAN (57.85%), (c) DSAN (64.01%), (d) TSTnet (61.65%), (e) CLDA (59.10%), (f) DDAN-JCA (59.01%), (g) MSDA (61.42%), (h) DHSNet (73.50%), (i) ground truth, and (j) pseudo-color image of Loukia.

F. Design of CFAAC Block

To design a module that can enhance the capacity for domain-invariant feature capturing and representation, we explored eight different implementations of the CFAAC block as depicted in Fig. 10. To amplify the weight calculated by the similarity of key and query, we removed the use of the softmax function. Correspondingly, to compensate for the loss of nonlinear representation capability in the CFAAC block resulting from this approach, we incorporated the GELU activation function into the block. The main difference between the eight implementations lies in the position of the activation function.

Repeated experiments on Pavia dataset (patch size = 9 and $\lambda_{\mathrm{LMMD}} = \lambda_{\mathrm{ST}} = 1$) yielded the results shown in Table VIII. Specifically, applying the GELU activation function after both the key and value branches achieved the best performance, with OA reaching 91.90%, AA reaching 91.65%, and the

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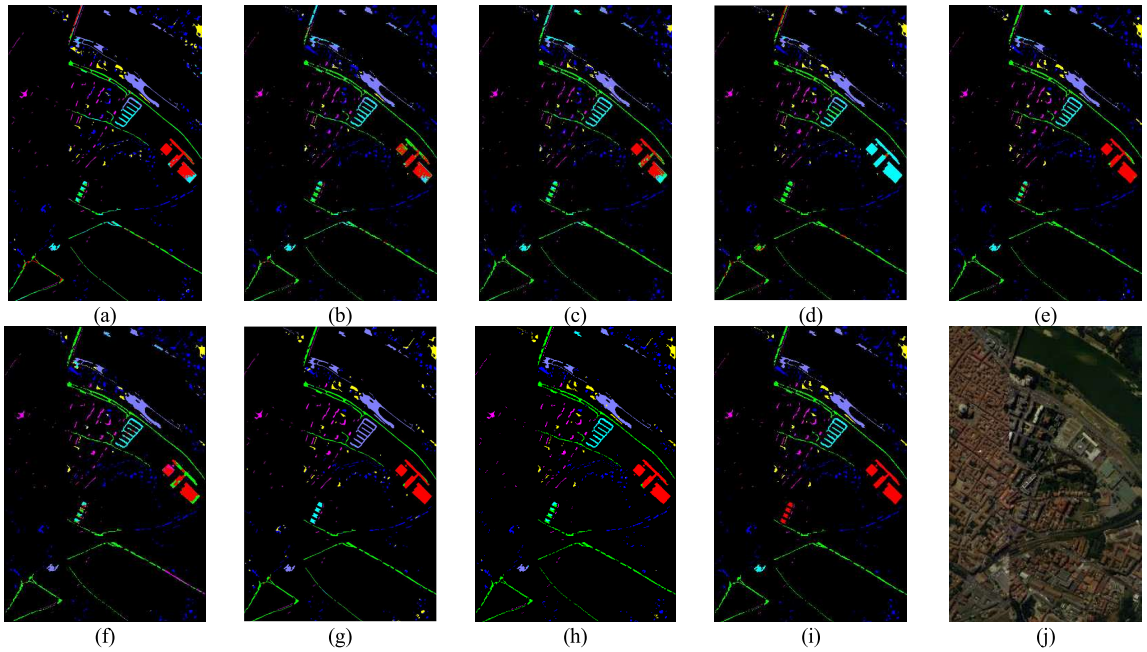


Fig. 7. Visualization of classification maps by different methods on Pavia dataset, including: (a) SpectralFormer (78.63%), (b) DAAN (78.51%), (c) DSAN (81.28%), (d) TSTnet (71.79%), (e) CLDA (93.13%), (f) DDAN-JCA (83.67%), (g) MSDA (88.30%), (h) DHSNet (93.55%), (i) ground truth, and (j) pseudo-color image of Pavia Center.

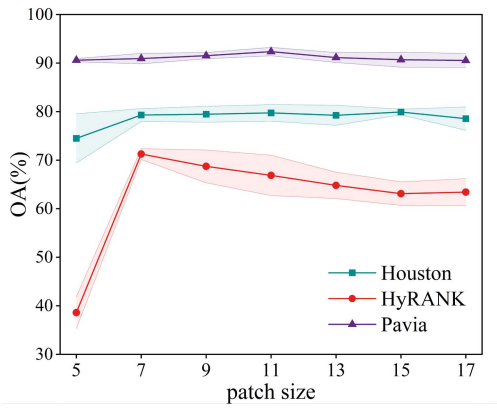


Fig. 8. OA (%) of different patch sizes on the three datasets. The shaded area represents the standard deviation.

TABLE VIII
PERFORMANCE OF DIFFERENT BLOCK DESIGNS

Design	a	b	c	d
OA (%)	91.60±0.97	91.77±0.66	91.47±0.95	91.32±0.92
AA (%)	91.34±1.17	91.09±0.90	91.15±0.97	90.65±1.09
Kappa ×100	89.89±1.17	90.09±0.79	89.73±1.15	89.54±1.10
Design	e	f	g	h
OA (%)	91.40±0.85	91.71±0.75	91.40±0.85	**91.90±0.56**
AA (%)	90.92±1.05	91.29±0.97	90.92±1.05	**91.65±1.12**
Kappa ×100	89.65±1.02	90.02±0.90	89.65±1.02	**90.26±0.68**

Kappa coefficient reaching 90.26. Except for the default design of the CFAAC block (group h), only the variant that applies the activation function solely after the key vector (group b, OA = 91.77%) and the variant that applies it after both the key and the depth-wise convolution (group f, OA = 91.71%) performanced better than the variant without any activation function (group a, OA = 91.60%). This suggests that the activation function does not universally enhance the model's representational capacity, whereas activating the key vector may play a critical role in feature extraction.

TABLE IX
COMPUTATIONAL COST COMPARISON OF DIFFERENT METHODS ON HOUSTON DATASET

Method	Training time per epoch (s)	FLOPs (G)	Number of trainable parameters
DAAN	1.20	18.13	34.92M
DSAN	1.00	16.03	24.89M
SpectralFormer	**0.24**	**0.72**	0.35M
TSTnet	0.36	3.02	9.81M
CLDA	0.74	27.16	**0.18M**
DDAN-JCA	0.92	13.97	1.13M
MSDA	1.38	15.41	1.14M
DHSNet	0.26	1.85	6.21M

G. Computational Efficiency Analysis

To provide a comprehensive understanding of the tradeoff between performance and computational cost, we conducted an analysis of the training time, inference time, and model complexity of DHSNet compared with representative baseline methods. All models were evaluated under the same experimental settings as in the main comparison experiments, using identical hardware configurations and batch sizes to ensure fairness.

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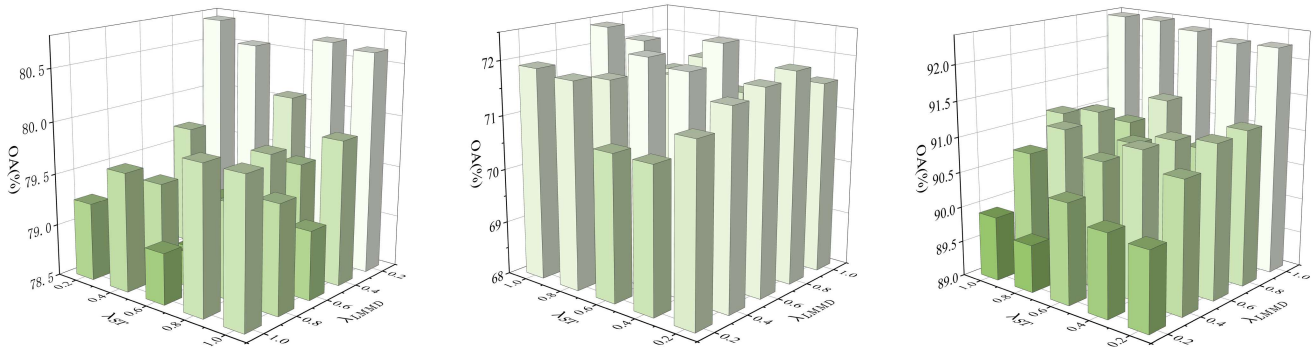


Fig. 9. OA (%) of different combinations of λ_{LMMD} and λ_{ST}. (a) Houston. (b) HyRANK. (c) Pavia.

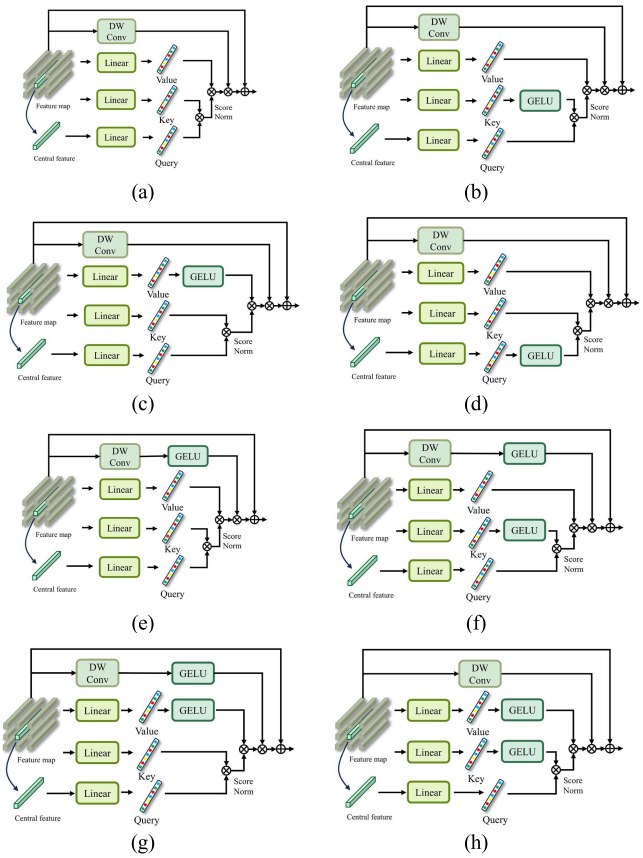


Fig. 10. Different implementations of the CFAAC block, including: (a) without activation function; (b) activation after key; (c) activation after value; (d) activation after query; (e) activation after deep-wise convolution; (f) activation after both deep-wise convolution and key; (g) activation after both deep-wise convolution and value; and (h) default design of the CFAAC block used in DHSNet, with activation after key and value.

For each method, we recorded the average training time per epoch and the total number of trainable parameters. This analysis aims to quantify the overhead introduced by the dual head architecture and the central feature attention mechanism in DHSNet. The results are summarized in Table IX, where the best performance is bolded, and the second-best performance is underlined. As shown, DHSNet achieves the second lowest FLOPs (1.85 G) and average training time per epoch (0.26 s), slightly behind SpectralFormer with 0.72 G FLOPs and 0.24 s. In contrast, as shown in Table IV, SpectralFormer attains 71.68% OA on Houston dataset, whereas DHSNet reaches 80.23%, representing a significant improvement of 8.55%. This further demonstrates that, while DHSNet introduces a moderate increase in computational requirements, it maintains a favorable balance between efficiency and performance.

This evaluation confirms that DHSNet is not only effective in improving cross-scene classification accuracy but also remains computationally feasible for practical use in real-world scenarios.

V. Conclusion

In this article, we propose an efficient network, DHSNet, for HSI cross-scene classification. The framework integrates class-wise DA and self-training to collaboratively achieve fine-grained feature alignment and mitigate the model's training bias toward the SD. In addition, a novel CFAAC block is incorporated into our feature extractor, demonstrating its effectiveness in extracting cross-scene-invariant features and showing potential for other HSI cross-scene classification methods. Comprehensive experiments and analyses on three cross-scene HSI datasets indicate that DHSNet outperforms state-of-the-art methods. In future work, we will concentrate on addressing the category classification accuracy gap, exploring more detailed improvements in DA and self-training.

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